CONTINUED

Discussion questions

1 Why do you think that controlling the flow of electricity through a computer is referred to as Boolean logic?

2 What do you think is the purpose of a logic gate?

3 Do you think a computer would be able to operate without the use of logic gates? Why?

10.1 The role of logic gates

You learnt in Chapter 1 how data is converted from analogue data to digital data so that it can be processed by a computer. Therefore, you understand that the data that is processed by a computer has two values, 0 and 1. A computer is provided with electricity to function and it needs a way to change this electricity from high voltage (1) to low voltage (0) and vice versa, to be able to represent each binary value that is required. It can do this through the use of logic gates. The reason why this chapter is called Boolean logic is because a Boolean value only has two different states, e.g. true or false, or in a logic gate's case 0 or 1.

Logic gates can have several inputs and depending on the type of logic the gate uses, it will change these inputs in one single output. All the logic gates that you will learn about will have a maximum of two input values.

A computer uses thousands of logic gates to output the high or low voltage that is needed to process and store the large quantity of data that it is provided with on a daily basis. The logic gates that you will learn about are the NOT, AND, OR, NAND, NOR and XOR gates.

10.2 The NOT logic gate

The NOT gate is the simplest logic gate. This logic gate has only one input and one output. It is used to reverse the input into the gate to output the opposite value. So, if high voltage (1) is input, the output will be low voltage (0). If low voltage (0) is input, high voltage (1) is output.

Figure 10.2 shows how to draw a NOT logic gate:

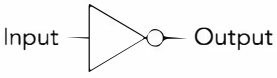


Figure 10.2: A NOT logic gate

A **truth table** can be used to represent the logic of a logic gate. It shows all the combinations that can be used as input values and what the output value will be for those input values. When doing this it is easier to give the input and output a character as a label, for example, the input can be labelled A and the output can be labelled X. This is a completed truth table for a NOT logic gate:

KEY WORD

truth table: a method to show all the different outcomes of an electrical circuit or system, dependent on the inputs it receives.

A	X
1	0
0	1

A **logic expression** can be used to represent the logic of a logic gate. It shows the logic as a mathematical equation. The logic expression for a NOT gate is:

$$X = \text{NOT } A$$

When you are doing further research and practice with Boolean logic, you may find that there are other symbols and notations that can be used in logic expressions. An alternative notation for a NOT gate is:

$$X = \overline{A}$$

The line over the top of the character A represents it is NOT A. An interesting point of knowledge to impress your friends with is that the line is called vinculum.

KEY WORD

logic expression: a mathematical representation that is used to represent an electrical circuit or system.

Questions

1 Draw a NOT logic gate.

2 Complete the truth table for the NOT logic gate. A is the input and X is the output:

A	X
1	
0	

3 Write the logic expression for a NOT logic gate.

10.3 The AND logic gate

The AND logic gate has two inputs and one output. The only time the output is 1 is when both the inputs are 1. A fun way to think about logic gates is that they are a doorway that has a guard. You and your friend are the inputs and you need the output to be high voltage (1) for the guard to allow you to be able to pass through the doorway. The criteria the guard has set for you to be able to get through the doorway for an AND logic gate is that you and your friend must both have a ticket. If neither of you has a ticket, or only one of you has a ticket, then you will not be allowed through the doorway.

Figure 10.3 shows how to draw an AND logic gate:

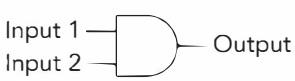


Figure 10.3: An AND logic gate

This time input 1 is labelled A, input 2 is labelled B and the output is labelled X.
This is a completed truth table for an AND gate:

A	B	X
0	0	**0**
0	1	**0**
1	0	**0**
1	1	**1**

The logic expression for an AND logic gate is:

$$X = A \text{ AND } B$$

An alternative notation for an AND gate is:

$$X = A.B$$

The dot symbol (.) between the A and the B represents the AND in this expression.

Questions

4 Draw an AND logic gate.

5 Complete the truth table for the AND logic gate:

A	B	X
0	0	
0	1	
1	0	
1	1	

6 Write the logic expression for an AND logic gate.

ACTIVITY 10.1

Look at the truth table that you have completed for question 5. From the results of the logic operations, do you think that an AND gate performs an addition, subtraction, multiplication or division mathematical operation? Why?

Peer Assessment

Discuss your thoughts from the activity with a partner and see if you agree on the mathematical operation that the AND gate performs. If you don't agree, discuss each of your reasons for your choice and work out who is correct.

Did you both arrive at this answer in the same way? Discuss how you worked out your answer.

10.4 The OR logic gate

The OR logic gate has two inputs and one output. The output is 1 when either or both the inputs are 1. The criteria the guard has set for you this time to be able to get through the doorway for an OR logic gate is that you *or* your friend must have a ticket, but both of you don't need one. If neither of you has a ticket, then you will still not be allowed through the doorway.

Figure 10.4 shows how to draw an OR logic gate:

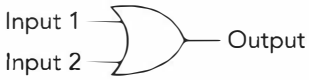


Figure 10.4: An OR logic gate

Input 1 is labelled A, input 2 is labelled B and the output is labelled X.
This is a completed truth table for an OR gate:

A	B	X
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**1**

The logic expression for an OR logic gate is:

$$X = A \text{ OR } B$$

An alternative notation for an OR gate is:

$$X = A + B$$

The plus symbol (+) between the A and the B represents the OR in this expression.

Questions

7 Draw an OR logic gate.

8 Complete the truth table for the OR logic gate:

A	B	X
0	0	
0	1	
1	0	
1	1	

9 Write the logic expression for an OR logic gate.

ACTIVITY 10.2

Look at the truth table that you have completed for question 8. From the results of the logic operations, do you think that an OR gate performs an addition, subtraction, multiplication or division mathematical operation? Why?

Peer Assessment

Discuss your thoughts from the activity with a partner and see if you agree on the mathematical operation that the OR gate performs. If you don't agree, discuss each of your reasons for your choice and work out who is correct.

Did you both arrive at this answer in the same way? Discuss how you worked out your answer.

10.5 The NAND logic gate

The NAND logic gate has two inputs and one output. The output is 1 when either input is 0 or one of the inputs are 1. If both inputs are 1, the output is 0. The criteria the guard has set for you this time to be able to get through the doorway for a NAND logic gate is that if both you and your friend are wearing trainers, you will not be allowed through the doorway. If neither of you, or only one of you is wearing trainers, then you will be allowed through the doorway.

Figure 10.5 shows how to draw a NAND logic gate:

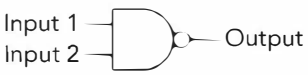


Figure 10.5: A NAND logic gate

Input 1 is labelled A, input 2 is labelled B and the output is labelled X.
This is a completed truth table for a NAND gate:

A	B	X
0	0	**1**
0	1	**1**
1	0	**1**
1	1	**0**

The NAND logic gate is just a simplification of an AND gate followed by a NOT gate (Figure 10.6):



Figure 10.6: A NAND gate is a simplification of an AND gate and a NOT gate

If you look at the truth table for the NAND gate, you should notice that the outputs are the opposite of what they are in the AND truth table. This is because the NOT gate takes the output of the AND table as its input, then reverses it to the opposite value.

The logic expression for a NAND logic gate is:

$$X = A \text{ NAND } B$$

An alternative notation for a NAND gate is:

$$X = \overline{A.B}$$

The dot symbol (.) between the A and the B and then the line over both inputs and the dot, represents the NAND in this expression.

COMPUTER SCIENCE IN CONTEXT

You may remember that you studied how data is stored using solid-state storage in the hardware chapter. In Chapter 3, you learnt that NAND gate technology is one of the technologies used to create solid-state storage.

Questions

10 Draw a NAND logic gate.

11 Complete the truth table for the NAND logic gate:

A	B	X
0	0	
0	1	
1	0	
1	1	

12 Write the logic expression for a NAND logic gate.

10.6 The NOR logic gate

The NOR logic gate has two inputs and one output. The only time the output is 1 is when both inputs are 0. The criteria the guard has set for you this time to be able to get through the doorway for a NOR logic gate is that neither of you are wearing trainers. If either or both of you are wearing trainers, then you will not be allowed through the doorway.

Figure 10.7 shows how to draw a NOR logic gate:

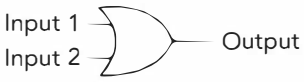


Figure 10.7: A NOR logic gate

Input 1 is labelled A, input 2 is labelled B and the output is labelled X.
This is a completed truth table for a NOR gate:

A	B	X
0	0	**1**
0	1	**0**
1	0	**0**
1	1	**0**

The NOR logic gate is just a simplification of an OR gate followed by a NOT gate (Figure 10.8):



Figure 10.8: A NOR gate is simplification of an OR gate and a NOT gate

If you look at the truth table for the NOR gate, you should notice that the outputs are the opposite of what they are in the OR truth table. This is because the NOT gate takes the output of the OR table as its input, then reverses it to the opposite value.

The logic expression for a NOR logic gate is:

$$X = A \text{ NOR } B$$

An alternative notation for a NOR gate is:

$$X = \overline{A+B}$$

The plus symbol (+) between the A and the B and then the line over both inputs and the +, represents the NOR in this expression.

Questions

13 Draw a NOR logic gate.

14 Complete the truth table for the NOR logic gate:

A	B	X
0	0	
0	1	
1	0	
1	1	

15 Write the logic expression for a NOR logic gate.

10.7 The XOR logic gate

The XOR logic gate is also known as the EOR gate. This is because it is known as the exclusive OR gate. It has two inputs and one output. The only time the output is 1 is when one of the inputs is 1. If both inputs are 0 or 1, then the output is 0. The criteria the guard has set for you this time to be able to get through the doorway for a XOR logic gate is that one of you is wearing trainers. If neither or both of you are wearing trainers, then you will not be allowed through the doorway.

Figure 10.9 shows how to draw a XOR logic gate:

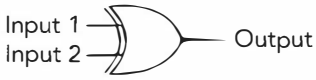


Figure 10.9: An XOR logic gate

Input 1 is labelled A, input 2 is labelled B and the output is labelled X.
This is a completed truth table for an XOR gate:

A	B	X
0	0	**0**
0	1	**1**
1	0	**1**
1	1	**0**

If you look at the truth table for the XOR gate, you should notice that the outputs are very similar to what they are in the OR truth table. The only difference is that if both inputs are 1, the output is 0.

The logic expression for a XOR logic gate is:

$$X = A \text{ XOR } B$$

An alternative notation for an XOR gate is:

$$X = A \oplus B$$

The plus symbol (+) within a circle, between the A and the B, represents the XOR in this expression.

Questions

16 Draw an XOR logic gate.

17 Complete the truth table for the XOR logic gate:

A	B	X
0	0	
0	1	
1	0	
1	1	

18 Write the logic expression for an XOR logic gate.

SKILLS FOCUS 10.1

REPRESENTING LOGIC CIRCUITS

A logic circuit is a combination of several logic gates that are linked. Figure 10.10 shows a simple logic circuit:

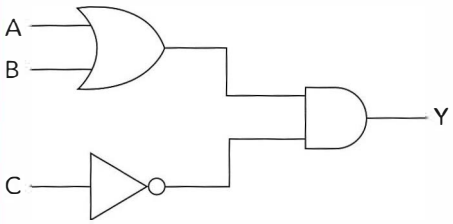


Figure 10.10: A simple logic circuit

There are three different inputs to this logic circuit: A, B and C. There is one output from this logic circuit that is labelled Y. There are three different logic gates in this logic circuit: NOT, OR and AND.

You need to be able to complete a truth table for a logic circuit, so let's learn how to do that.

First you need a truth table to complete. As this logic circuit has three inputs, so there can be eight different combinations to consider:

A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

If you look at the values in the truth table, what do you notice about the order of each combination? If you think about each row as a 3-bit binary value, you should notice that they increment from 0 to 7 in denary. For example, 000 = 0, 001 = 1, 010 = 2, 011 = 3, etc.

In the logic circuit, as the output from the OR gate and the NOT gate become the inputs for the AND gate, it helps to give these a label so that it is easier to track values for them. In this example, Figure 10.11, the labels D and E are used.

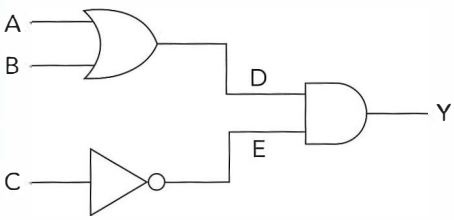


Figure 10.11: A simple logic circuit with labels D and E added

CONTINUED

You can add these labels into the truth table to work out what these interim values are:

A	B	C	D	E	Y
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

You can now begin to fill in the values in the truth table. You need to complete the output for the OR gate in the column labelled D. You have learnt that the logic of an OR gate is that if either or both inputs are 1, the output is 1. You are just looking at the values in columns A and B for this, as these are the inputs to the OR gate. You can ignore what is in column C for now. This is the truth table with column D completed:

A	B	C	D	E	Y
0	0	0	**0**		
0	0	1	**0**		
0	1	0	**1**		
0	1	1	**1**		
1	0	0	**1**		
1	0	1	**1**		
1	1	0	**1**		
1	1	1	**1**		

You need to complete the output for the NOT gate in the column labelled E. You have learnt that the logic of a NOT gate is that it reverses the input value to be the opposite value. You are just looking at the values in column C for this, as this is the input to the NOT gate. You can ignore what is in the other columns for now. This is the truth table with column E completed:

A	B	C	D	E	Y
0	0	0	0	**1**	
0	0	1	0	**0**	
0	1	0	1	**1**	
0	1	1	1	**0**	
1	0	0	1	**1**	
1	0	1	1	**0**	
1	1	0	1	**1**	
1	1	1	1	**0**	

CONTINUED

You can now complete the final output column that is the output for the logic circuit, this is column Y. You have learnt that the logic of an AND gate is that the only time the output is 1 is when both inputs are 1. You are just looking at columns D and E for this, as they are the inputs to the AND gate. This is the truth table with the final output column completed:

A	B	C	D	E	Y
0	0	0	0	1	**0**
0	0	1	0	0	**0**
0	1	0	1	1	**1**
0	1	1	1	0	**0**
1	0	0	1	1	**1**
1	0	1	1	0	**0**
1	1	0	1	1	**1**
1	1	1	1	0	**0**

You now know how to complete a truth table for a logic circuit, excellent!

Questions

1 Complete the truth table for the given logic circuit in Figure 10.12:

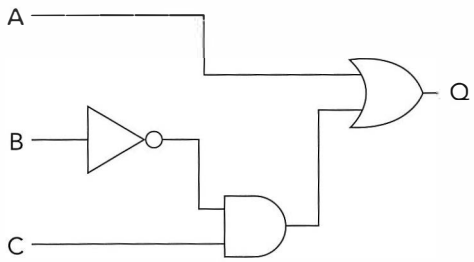


Figure 10.12: Complete the truth table for this logic circuit

A	B	C	Workspace	Q
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

CONTINUED

2 Complete the truth table for the given logic circuit in Figure 10.13:

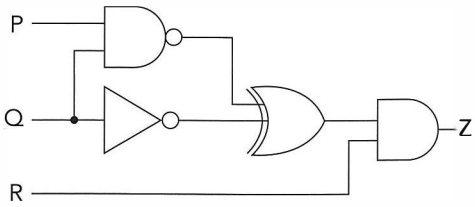


Figure 10.13: Complete the truth table for this logic circuit

P	Q	R	Workspace	Z
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

10.8 Logic expressions

You need to be able to write a logic expression for a logic circuit. The first thing you need to do to be able to write this is to break down the different parts of the logic circuit. Let's look at the logic circuit in Figure 10.14 as an example:

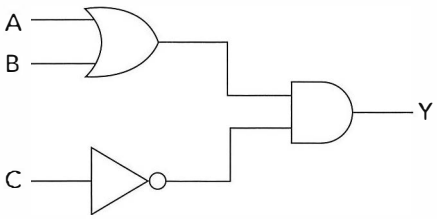


Figure 10.14: A simple logic circuit

The first part to the logic circuit is the OR gate. The logic expression for this part is A OR B. The second part of this logic circuit is NOT and the third part of this logic circuit in the AND gate.

So, the three parts to this logic circuit are:

A OR B

NOT C

AND

Questions

19 Write the logic expression for the given logic circuit:

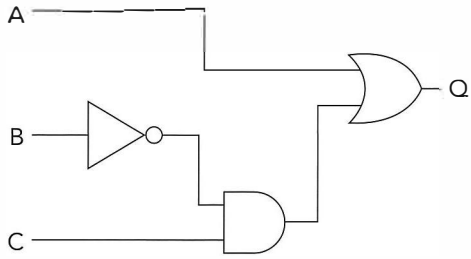


Figure 10.16: Find the logic expression for this circuit

20 Write the logic expression for the given logic circuit:

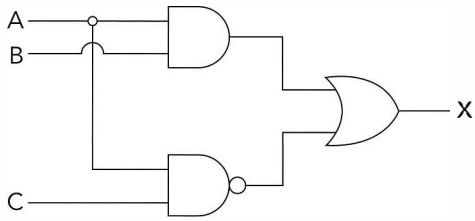


Figure 10.17: Find the logic expression for this circuit

10.9 Representing truth tables

You have learnt how to complete a truth table for a logic expression and a logic circuit. You now need to learn how to write a logic expression and draw a logic circuit from a truth table.

Let's start with a simple truth table that has two input values A and B:

A	B	X
0	0	**0**
0	1	**1**
1	0	**0**
1	1	**0**

If you look at this truth table, the first thing you should ask yourself if whether you know any single logic gates that would produce this truth table. It doesn't match any of the truth tables that you have learnt previously in this chapter for each of the logic gates, therefore you now know it must be a combination of at least two logic gates.

When you are creating a logic expression and a logic circuit from a truth table, you only need to know how to do this using three of the logic gates that you have learnt: the NOT gate, the AND gate and the OR gate.

Once you have looked at the truth table to see if it represents a single logic gate that you know, if you find it doesn't, you can move onto the next step which is finding out which gates will create that truth table.

Looking at the truth table again:

A	B	X
0	0	**0**
0	1	**1**
1	0	**0**
1	1	**0**

It has one output value that is a 1, so this is what you need to concentrate on. You need to use an AND gate to create this output. At present an input value of A = 0 and B = 1 will not result in an output of 1 using an AND gate. You should remember that for the output to be 1 for an AND gate, both inputs need to be 1. Therefore, you need to reverse input A, which is 0, to make it 1. You should remember that to reverse the input value to create the opposite value as an output, you should use a NOT gate. That makes the logic expression for this truth table:

X = (NOT A AND B)

You can then draw a logic circuit for this (Figure 10.18):

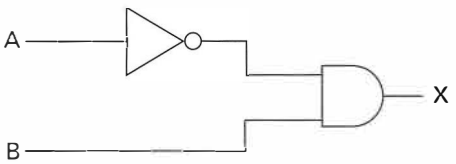


Figure 10.18: The logic circuit for logic expression X = (NOT A AND B)

Let's look at another truth table:

A	B	X
0	0	**0**
0	1	**1**
1	0	**0**
1	1	**1**

This time the truth table has two output values that are 1, so you need to focus on both of those. You have the same situation again with the first output of 1, so you already know that this part is NOT A AND B.

A	B	X
0	0	**0**
0	1	**1**
1	0	**0**
1	1	**1**

← NOT A AND B

You now need to look at the second output value of 1. You need to think whether an input of 1 for A and an input of 1 for B will create an output of 1 if an AND gate is used. You should recognise that it will. This means that this part is A AND B.

A	B	X	
0	0	**0**	
0	1	**1**	← NOT A AND B
1	0	**0**	
1	1	**1**	← A AND B

Now that you have worked out what the part of the logic expression is for each output value of 1, you need to join these together using an OR gate. That makes the logic expression for this truth table:

X = (NOT A AND B) OR (A AND B)

You can then draw a logic circuit for this (Figure 10.19):

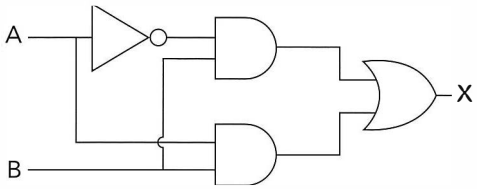


Figure 10.19: The logic circuit for logic expression X = (NOT A AND B) OR (A AND B)

Let's now look at a truth table that has three input values A, B and C:

A	B	C	X
0	0	0	**0**
0	0	1	**1**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**1**
1	1	0	**0**
1	1	1	**1**

This makes the process more complex and there are several methods that you can use to work out the logic expression. Each method may generate a slightly different logic expression, but they all should produce the same truth table. It is the structure of them that will change slightly, but the logic of them will ultimately produce the same result.

The first method you can use is a method called sum of products. In this method you need to create a sum of products expression for each row in the table that has an output value of 1. In a sum of products statement, you need to look at each input value. If it is 1, then in the expression it will be, for example, A. If it is 0, then in the expression it will be, for example, NOT A. Each input is then summed, this is done using the AND gate.

Let's look at the first row with an output of 1 in the table:

A	B	C	X
0	0	0	**0**
0	0	1	**1**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**1**
1	1	0	**0**
1	1	1	**1**

In this row:

- A is 0 so that's NOT A.
- B is 0 so that's NOT B.
- C is 1 so that's C.

These are then joined using AND gates making:

NOT A AND NOT B AND C

You need to follow this process for each row in the truth table that has an output of 1. You should see this result when you do:

A	B	C	X	
0	0	0	**0**	
0	0	1	**1**	← NOT A AND NOT B AND C
0	1	0	**0**	
0	1	1	**0**	
1	0	0	**0**	
1	0	1	**1**	← A AND NOT B AND C
1	1	0	**0**	
1	1	1	**1**	← A AND B AND C

Each of the sum of product expressions is then joined together with an OR gate creating the final expression:

X = (NOT A AND NOT B AND C) OR (A AND NOT B AND C) OR (A AND B AND C)

A person would normally look to then simplify this statement, but you do not need to understand how to do this. You could research how to do this though if you wanted to give yourself a challenge.

The second method you could use looks for the constant values in each row where the output is 1.

Let's look at another truth table:

A	B	C	X
0	0	0	**0**
0	0	1	**1**
0	1	0	**0**
0	1	1	**1**
1	0	0	**0**
1	0	1	**0**
1	1	0	**1**
1	1	1	**1**

Again, you look at the rows that have an output of 1. If you look at the first row with an output of 1, the input values are A = 0, B = 0, C = 1. You need to look to see if any of these values are constant. This means, are there any values that are the same between the inputs. You should notice that the input values for A and B are the same, 0 and 0.

You now need to look whether inputs of 0 and 0 for A and B always result in an output of 1 in each row in the table. If they do, this means that the expression for this row is:

NOT A AND NOT B

If they don't always result in an output of 1, then you use the sum of products expression for the row instead. If you look at row 1 in the table, you should notice that A and B both have an input of 0, but that the output is 0 and not 1. Therefore, you need to use the sum of products expression for row 2 instead, which is:

NOT A AND NOT B AND C

This is because we need to apply the rule you learnt earlier about reversing the inputs to make sure that the values will give an output of 1 for an AND gate.

If you look at the next row that has an output of 1, you should see that the inputs are A = 0, B = 1, C = 1. This time it is B and C that have the constant value as they are both the same. You now need to look at whether all rows in the table where B and C have the value 1, the result is an output of 1. You should be able to notice that the only other place in the table where the inputs are both 1 is row 8, the final row. This does result in an output of 1 as well, so this means that the expression for this row is:

A AND C

If you look at the final two rows in the table, they both that have an output of 1. If you have two outputs of 1 together like this, you can treat them as a pair. You apply the same method to see if there are any values that are constant. In row 7 the values are A = 1, B = 1 and C = 0, and in row 8 the values are A = 1, B = 1 and C = 1. You should notice that the A and B inputs are constant in both of these rows. You now look to see if there are any other instances of A and B both having an input of one in the table to see if this results in an output of 1 also. There are no other instances, so this means the expression for these two rows is:

A AND B

These three expressions are then joined together again with an OR gate, giving a final expression of:

X = (NOT A AND NOT B AND C) OR (A AND C) OR (A AND B)

If you work out the statement using the sum of products method, you will get:

X = (NOT A AND NOT B AND C) OR (NOT A AND B AND C) OR
(A AND B AND NOT C) OR (A AND B AND C)

The two statements represent the same truth table, but you can see that the second method creates a shorter statement, which can be helpful if there are lots of outputs of 1 to be considered.

Question

21 Look at the given truth table:

A	B	X
0	0	1
0	1	1
1	0	0
1	1	0

a Write a logic expression to represent the given truth table.

b Draw a logic circuit to represent the given truth table.

10.10 Representing logic expressions

You need to be able to look at a logic expression and create a logic circuit and a truth table from the expression.

Let's look at a simple logic expression:

Z = (R AND S) OR (R AND T)

You should notice that this logic expression has:

- 3 input values: R, S and T
- an output value of Z
- uses the logic gates AND and OR.

If you look at the brackets you can see that the expression has two main parts that are joined together by an OR gate. You should learn from this that the OR gate will be the final gate in the logic circuit. You can the draw this part of the circuit (Figure 10.20):

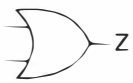


Figure 10.20: The logic circuit for the expression Z = (something) OR (something)

You the need to look at the parts of the expression that are in the brackets. Each bracketed section is an input into the OR gate that you have drawn. This means that the first input to the OR is the output of R AND S (Figure 10.21):

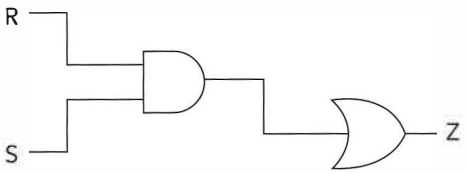


Figure 10.21: The logic circuit for the expression Z = (R AND S) OR (something)